

Design and Implementation of a Teaching System for Visual Stimulation and Recording of Single Unit and Mass Signals

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Introduction

Biosignals can be measured, monitored, and analyzed in order to provide information about physiological functions and health statuses. These signals are useful for different types of applications, e.g. evoked potentials (EP). Commercial biosignal data acquisition and logging systems are frequently very expensive, developed via closed-source software and hardware, limited in functionality, difficult to cure defects, difficult to add new features or not documented in detail. In this poster, a budget priced biosignal data acquisition, logging and visual stimulation system has been developed and implemented. The system has been designed to support and teach biomedical engineering students the in-vivo recording of neuronal signals, e.g. visually evoked potentials (VEP) or spike activities. The system helps in this context by allowing students to measure brain signals data in a flexible and interactive way from anaesthetized flies. Moreover, the system provides logging features with various options.

Methods

Mechanical parts

Handles, stands, etc. have been used to enable positioning of the electrodes and the fly. Moreover a faradic cage is used to avoid noise

Pre amplifier

A pre-amplifier manufactured by TREC/Gießen was used with self-designed electrodes

National Instruments USB-6211 multifunction I/O device

It has been used as a DAQ system to acquire data, save it as MAT-Files and to control the LED. LabVIEW [1] was used as programming language

Electronic-box

The electronic-box has been designed and assembled by the authors. It enables to amplify and filter signals. Moreover it provide controlling the LED with or without a DAQ

LED-Cannon

The RGB LED is controlled from other devices. The duration of the pulses, frequency and the intensity are adjustable

Photodiode

It is used to detect the stimuli of the LED. The signal will be used to compute peristimulus time histograms which are needed for illustration of evoked potentials and for comparison with VEP

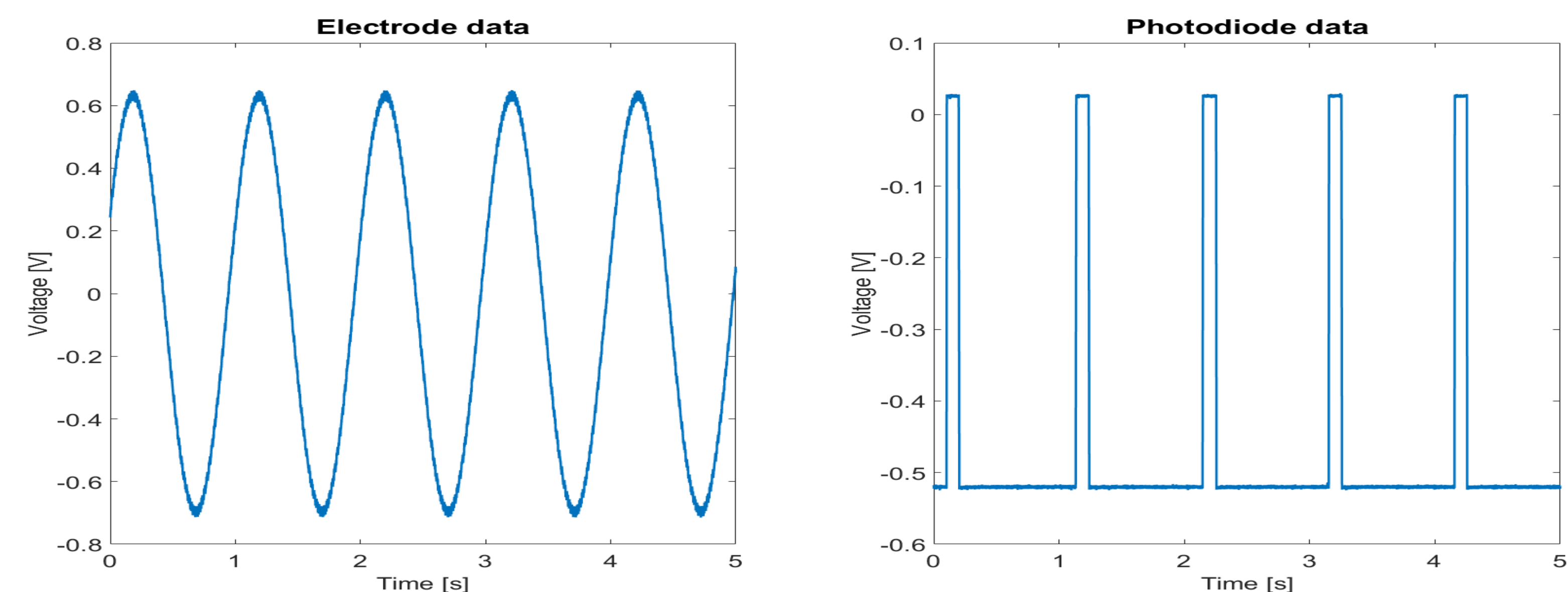


Figure 2: To test the acquisition system two signals were recorded. A sinus signal was fed into the electrodes with a function generator with the frequency of 1 (left). A stimulus with frequency of 1 was recorded with the Photodiode (right).

Results

- Controlling the LED via software and electronic-box is working
- Time problems by saving the signals with the LabVIEW software for high sampling rates (> 20 kHz)
- Detection of action potentials and visual evoked potentials is currently available with the Biopac system as DAQ system. The visual evoked potentials were recorded on the surface of the fly eye
- Amplification and filtering with the electronic-box is working

Discussion and Conclusion

- A biosignal data acquisition and logging system has been developed and implemented using LabVIEW programming language
- Implementing a own DAQ setup with LabVIEW is difficult for high sampling rates
- In the future the time problems by saving the data will be fixed
- The system helps biomedical engineering students to understand biosignal recording
- In the future the system will be used to detect single unit activity with the microelectrode manipulator using special fibre micro-electrodes

Acknowledgements

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References

[1] Nasser, K. and Namjin, K., 2005. Digital Signal Processing System-Level Design Using LabVIEW. Elsevier.

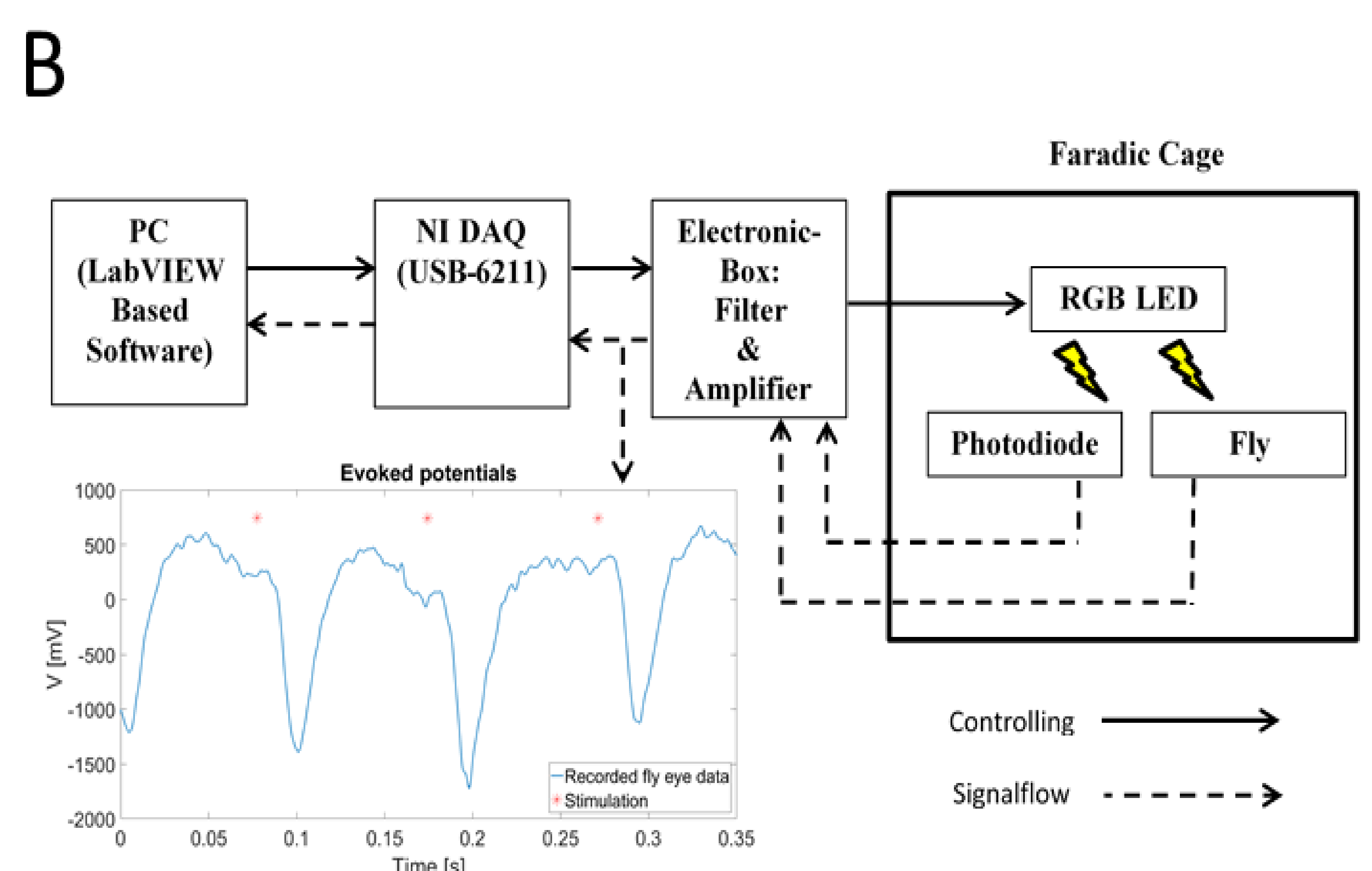
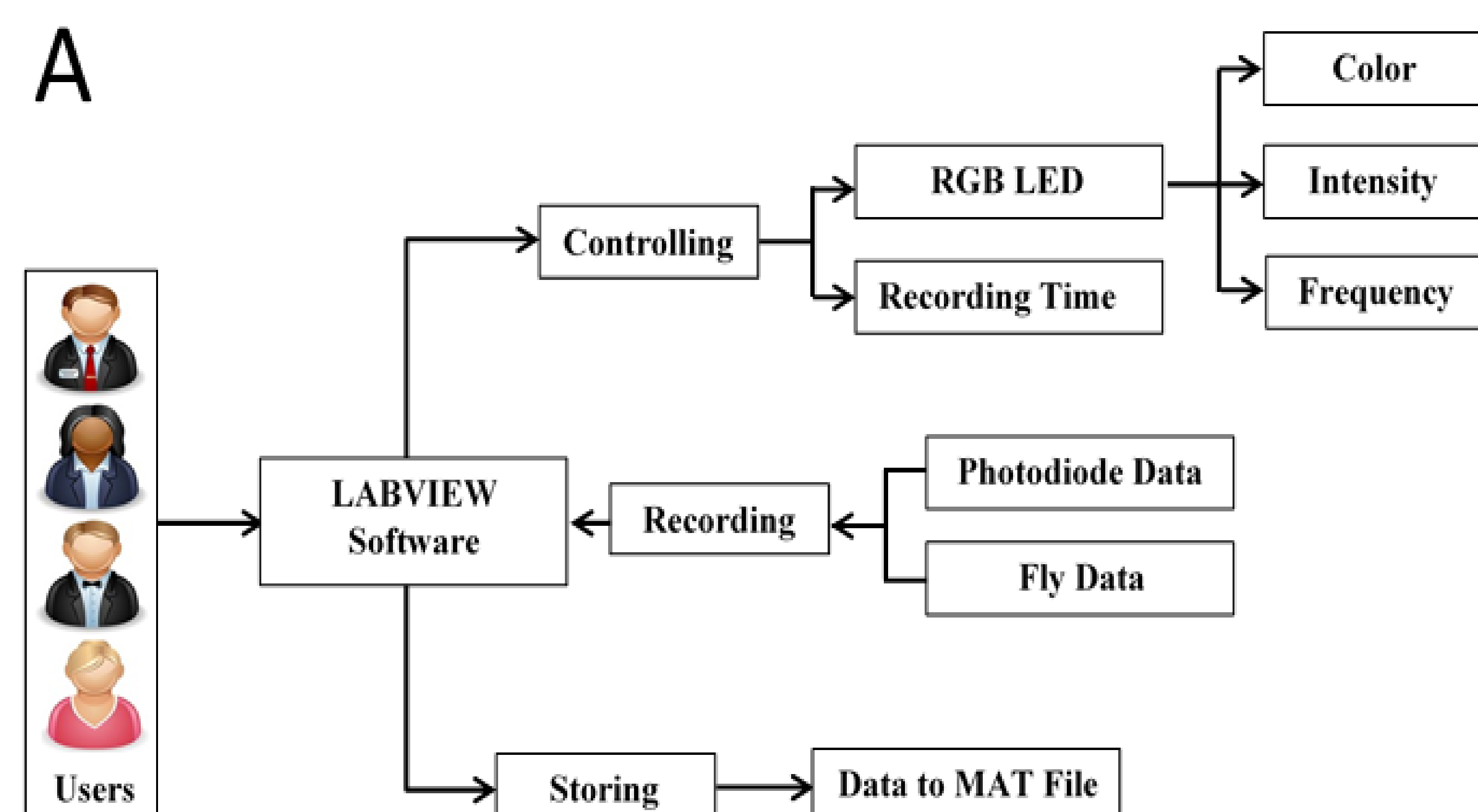


Figure 1: **A** illustration of the software part, **B** illustration of the hardware part including a signal of evoked potentials. **Left** The user can interact with the software which was implemented with LabVIEW. In the GUI the parameters for the RGB LED, color, intensity and frequency can be changed immediately. Moreover the recording time and the sample rate can also be changed online. The photodiode and fly eye responses will be recorded and then saved in a MAT file. **Right** The controlling flow show how each hardware part is controlled from a other device. At the end the RGB LED emitted light photons which are detected from the photodiode and the fly eye. The signal graph show how the recoded signal from the surface of the fly eye looks like. Red stars marks time point of stimulation.